

$$\text{REGULAR}_{\text{TM}} = \{ \langle M \rangle \mid M \text{ is a TM and } L(M) \text{ is regular} \}$$

$$A_{\text{TM}} \leq \text{REGULAR}_{\text{TM}}$$

$\langle M, w \rangle$ $\langle M_1 \rangle$

I am going to create special TM M_1 ,

$M_1 =$ "on input $\langle x \rangle$, m and w are encoded in this TM"

1. If $x \in 0^n 1^n$, accept
2. If $x \notin 0^n 1^n$, run M on w and accept x if M accepts w .

$$M \text{ accepts } w \Rightarrow L(M_1) = \Sigma^*$$

$L(M_1)$ is regular.

$$M \text{ does not accept } w \Rightarrow L(M_1) = \{ 0^n 1^n \mid n \geq 0 \}$$

$\nearrow$
 not a regular language.

Let's create the decider for A_{TM} .

Assume R is the decider for $REGULAR_{TM}$.

Let's create the decider S for A_{TM}

$S =$ "on input $\langle M, w \rangle$:

1. construct TM M_1 by encoding $\langle M, w \rangle$ in its construction
2. Run R on $\langle M_1 \rangle$
3. If R accept $\langle M_1 \rangle$, accept
If R rejects $\langle M_1 \rangle$, reject

Here, TM S is a decider for A_{TM} . Obviously, this should not be possible.

$\therefore REGULAR_{TM}$ is undecidable.

$$EQ_{TM} = \left\{ \langle M_1, M_2 \rangle \mid M_1, M_2 \text{ are TMs and } L(M_1) = L(M_2) \right\}$$

$$A_{TM} \leq EQ_{TM}$$

$\langle M, w \rangle$ $\langle M_1, M_2 \rangle$

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Harder to do

$$E_{TM} \leq EQ_{TM}$$

$\langle M \rangle$ $\langle M_1, M_2 \rangle$

Assume EQ_{TM} is decidable, and TM R decides it.

Then we construct TM S to decide E_{TM} .

$S =$ on input $\langle M \rangle$, where M is a TM

1. Run R on $\langle M, T \rangle$ where TM T does not accept any string.
2. If R accepts, accept
If R rejects, reject.

Q: what is $\overline{A_{TM}}$?

$$A_{TM} = \{ \langle M, w \rangle \mid M \text{ is a TM and } M \text{ accepts } w \}$$

$$\overline{A_{TM}} = \{ \langle M, w \rangle \mid M \text{ does not accept } w \}$$

Turing-unrecognizable languages

Any language that is not Turing-recognizable is called Turing-unrecognizable

Def

A language is co-Turing recognizable if it is the complement of a Turing recognizable language

$\overline{A_{TM}}$ is co-Turing recognizable

$\overline{\overline{A_{TM}}} = A_{TM}$, and we know that A_{TM} is Turing recognizable.

$\therefore \overline{A_{TM}}$ is co-Turing recognizable.

Theorem 4.22: A language is decidable if and only if it is both Turing recognizable and co-Turing recognizable.

Proof: Suppose A is recognized by TM M_1 and $\bar{A}$ is recognized by TM M_2 , then we construct TM M which is a decider for A .

$M =$ " on input w :

1. Run M_1 and M_2 on w in parallel.

2. If M_1 accept, accept;
If M_2 accept, reject.

$\overline{A_{TM}}$ is not Turing recognizable

Why?

we know that A_{TM} is undecidable

Theorem 4.22 says.

A is decidable $\Leftrightarrow A$ is Turing recognizable
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A is co-Turing recognizable

A is undecidable $\Leftrightarrow A$ is not Turing recognizable

or

A is not co-Turing recognizable.

Since I know that A_{TM} is undecidable
and A_{TM} is Turing recognizable

A_{TM} must be not co-Turing recognizable
in order for Theorem 4.22 to be
correct

